Chapter 7: Law of Large Numbers and the Central Limit Theorem

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October 11, 2026

CSAI2121 Probability and Statistics
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Learning Objectives

By the end of this lecture, you should be able to:

- Define the **sample mean** and derive its expectation and variance
- State the **Law of Large Numbers** (LLN) and explain its intuitive meaning
- State the **Central Limit Theorem** (CLT) and interpret its standardized form
- Distinguish clearly between what the LLN and CLT tell us
- Compute the **standard error** of a sample mean
- Use the CLT to perform approximate probability calculations
- Connect the LLN and CLT to Monte Carlo estimation, mini-batch learning, model evaluation, and ensemble methods

From One Observation to a Sample

Motivation: One Draw vs. Many Draws

- So far, we have mainly studied the distribution of a **single** random variable or random vector.
- In practice, we usually collect a **sample**:

$$X_1, X_2, \ldots, X_n.$$

- Examples:
  - responses from many users;
  - measurements from many sensors;
  - prediction results on many test examples;
  - repeated simulations of a probabilistic system.
- A central question in statistics is:

  What happens when we average many random observations?

Bridge to Statistical Inference

The behaviour of averages connects probability theory to parameter estimation, confidence intervals, hypothesis testing, and machine learning.

Recall: i.i.d. Random Variables

Definition

Random variables $X_1, X_2, \ldots, X_n$ are **independent and identically distributed** (i.i.d.) if:

1. they are mutually independent; and
2. they all follow the same probability distribution.

If the common distribution has mean μ and variance σ^2, then:

$$E[X_i] = \mu, \qquad \mathrm{Var}(X_i) = \sigma^2 \quad \text{for every } i.$$

- Independence: one observation does not reveal the outcome of another.
- Identically distributed: all observations come from the same population or data-generating process.

AI Interpretation

A randomly sampled training or test set is often modelled as i.i.d. observations from an underlying data distribution.

The Sample Mean

Definition

For observations $X_1, \ldots, X_n$, the **sample mean** is

$$\overline{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

- μ is the fixed but usually unknown **population mean**.
- $\overline{X}_n$ is a **random variable**, because its value depends on the random sample.
- Once a particular dataset is observed, the numerical value is written as

$$\overline{x} = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

Important Distinction

Different random samples generally produce different values of $\overline{X}_n$, even when they come from the same population.

Expectation of the Sample Mean

Suppose $X_1, \ldots, X_n$ are identically distributed with $E[X_i] = \mu$.

Using linearity of expectation:

$$\begin{aligned} E[\overline{X}_n] &= E\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] \\ &= \frac{1}{n}\sum_{i=1}^{n} E[X_i] \\ &= \mu. \end{aligned}$$

Interpretation

The sample mean is an **unbiased estimator** of the population mean:

$$E[\overline{X}_n] = \mu.$$

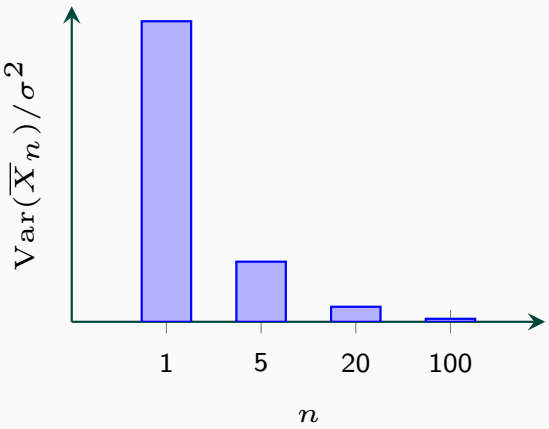
Across repeated samples, it is correct on average.

Independence is not required for this expectation result.

Variance of the Sample Mean

Suppose the observations are independent and $\mathrm{Var}(X_i) = \sigma^2$.

$$\begin{aligned}\mathrm{Var}(\overline{X}_n) &= \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) \\ &= \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) \\ &= \frac{1}{n^2}(n\sigma^2) = \frac{\sigma^2}{n}.\end{aligned}$$

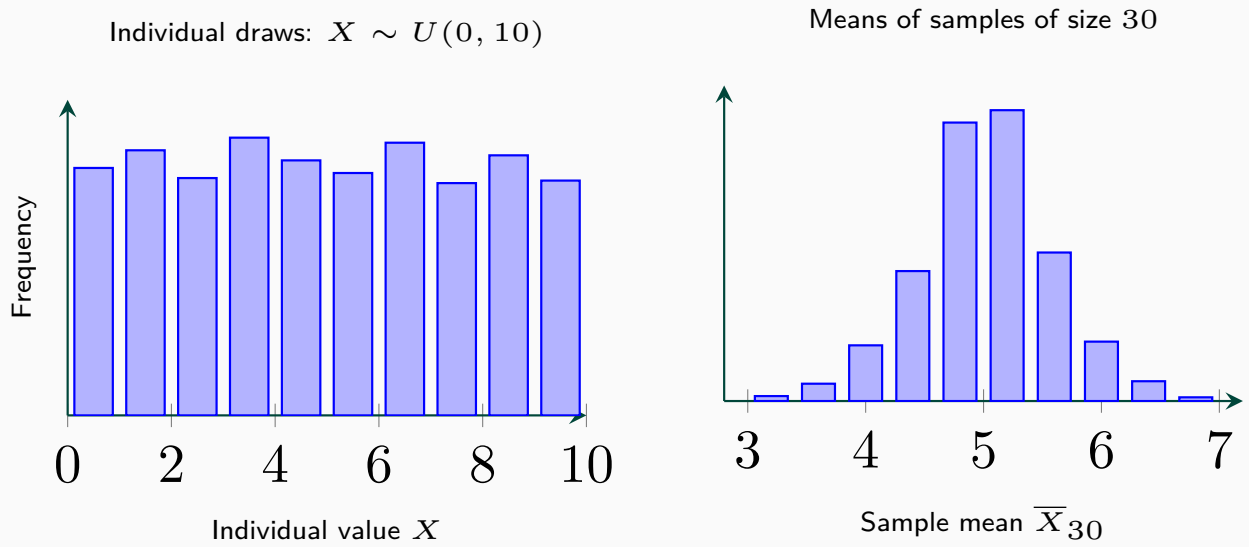


Key Result

$$\mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n}.$$

As the bar chart shows, more observations make the sample mean less variable and more concentrated around μ.

Individual Observations vs. Sample Means



- Both distributions have approximately the same centre: $\mu = 5$.
- The sample means are much **less spread out**.
- Their distribution is also more **bell-shaped**.

The Law of Large Numbers

Intuition: Averaging Reduces Randomness

Consider independent flips of a fair coin:

$$X_i = \begin{cases} 1, & \text{if flip } i \text{ is heads,} \\ 0, & \text{if flip } i \text{ is tails.} \end{cases}$$

Then:

$$E[X_i] = 0.5, \qquad \overline{X}_n = \frac{\text{number of heads}}{n}.$$

- One flip is completely unpredictable.
- The proportion of heads after ten flips can still vary greatly.
- After thousands of flips, the proportion is usually close to 0.5.

Main Intuition

Individual outcomes remain random, but averaging many independent outcomes tends to cancel positive and negative fluctuations.

Law of Large Numbers

Law of Large Numbers: Informal Statement

If $X_1, X_2, \ldots$ are i.i.d. with finite mean μ, then the sample mean approaches the population mean as the sample size increases:

$$\overline{X}_n \longrightarrow \mu \qquad \text{as } n \longrightarrow \infty.$$

A useful probabilistic statement is:

$$P\left(|\overline{X}_n - \mu| > \varepsilon\right) \longrightarrow 0 \qquad \text{for every } \varepsilon > 0.$$

- ε represents any chosen error tolerance.
- As n grows, it becomes increasingly unlikely that the sample mean differs from μ by more than ε.

Plain-Language Meaning

With sufficiently many observations, the sample average is likely to be close to the true population average.

Why Does the LLN Work?

Recall:

$$E[\overline{X}_n] = \mu, \qquad \mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n}.$$

Therefore:

$$\mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n} \longrightarrow 0.$$

Under the standard i.i.d. and finite-variance assumptions, Chebyshev's inequality gives:

$$P\left(|\overline{X}_n - \mu| \geq \varepsilon\right) \leq \frac{\sigma^2}{n\varepsilon^2} \longrightarrow 0.$$

Interpretation

The centre remains at μ, while the spread around μ becomes smaller and smaller. This concentration is the mathematical basis of the LLN.

Chebyshev's inequality is included only to show the idea; its proof is not required.

Worked Example: Repeated Coin Flips

For a fair coin:

$$\mu = 0.5, \qquad \sigma^2 = 0.25.$$

One simulated sequence gives:

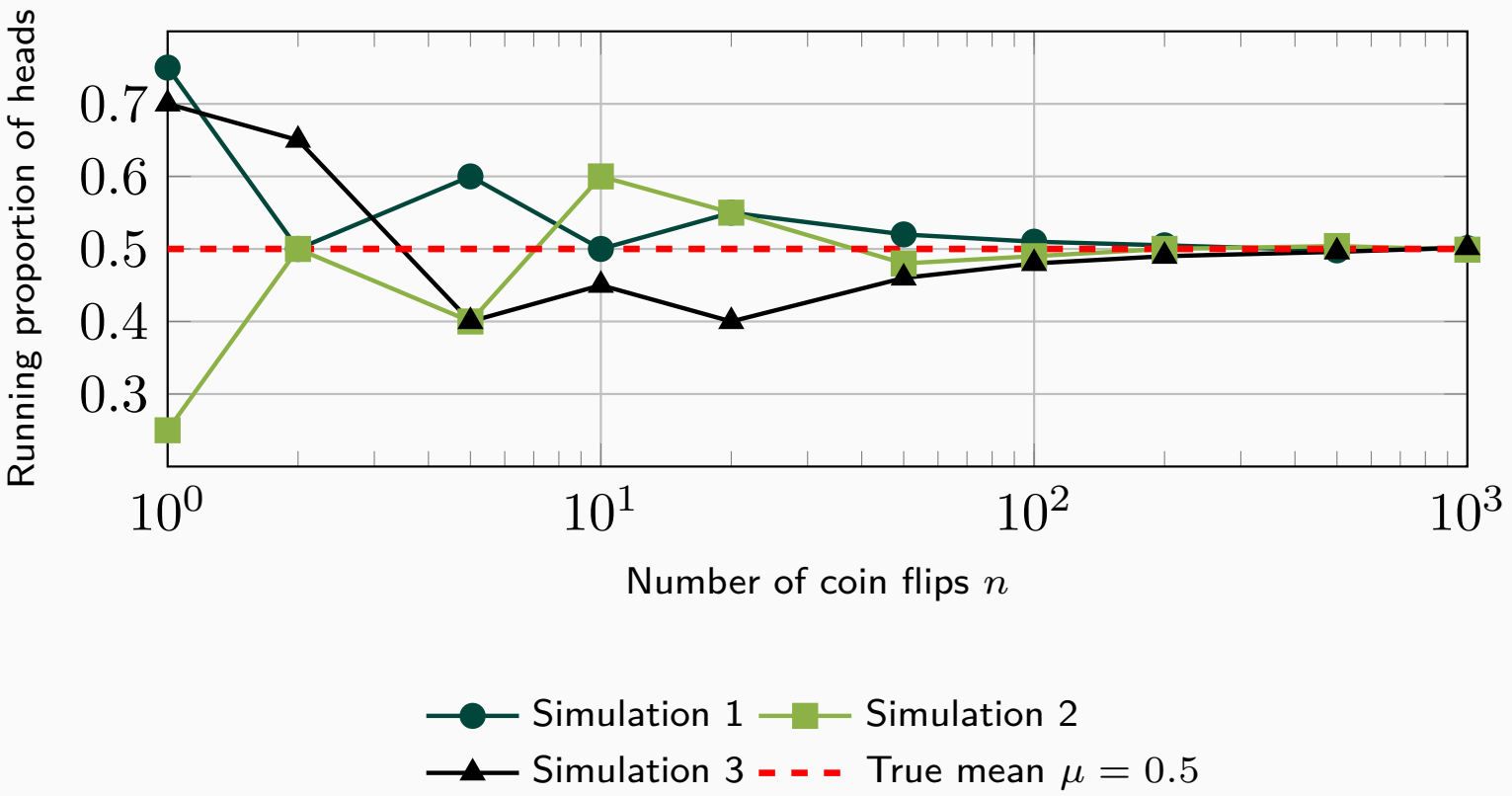
Number of flips n	Number of heads	Running average $\overline{x}_n$
1	1	1.000
10	6	0.600
50	27	0.540
200	101	0.505
1000	498	0.498

- The average may fluctuate greatly when n is small.
- It does not have to move steadily toward 0.5.
- The fluctuations generally become smaller as n increases.

Important

The LLN describes long-run concentration, not exact equality at any finite sample size.

Visualization: Running Averages Converge



- Different sequences follow different paths, but their running averages tend to settle near 0.5.

Common Misconception: The Gambler's Fallacy

Incorrect Reasoning

"The coin has landed heads five times in a row, so tails is now more likely in order to balance the results."

For independent fair-coin flips:

$$P(X_{n+1} = \text{heads} \mid X_1, \ldots, X_n) = 0.5.$$

- Previous flips do not change the probability of the next flip.
- The LLN does not provide a short-term correction mechanism.
- A long sequence can still contain unusual local patterns.
- The effect of an early imbalance becomes small because it is divided by an increasingly large n.

For example, if the first five flips are all heads, their contribution to the average is:

$$\frac{5}{n},$$

which approaches 0 as the total number of flips n grows.

AI Application: Monte Carlo Estimation

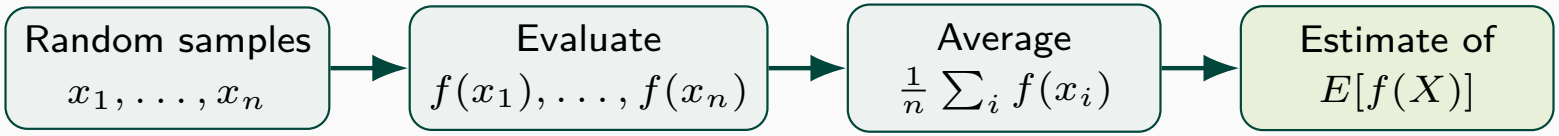
Suppose we want to calculate:

$$E[f(X)],$$

but the exact sum or integral is too difficult.

Monte Carlo estimation uses random samples:

$$x_1, \ldots, x_n \sim p(x), \qquad E[f(X)] \approx \frac{1}{n} \sum_{i=1}^{n} f(x_i).$$



Role of the LLN

As n increases, the Monte Carlo average converges to the true expectation under suitable conditions.

Examples include expected rewards in reinforcement learning and simulation-based probability estimates.

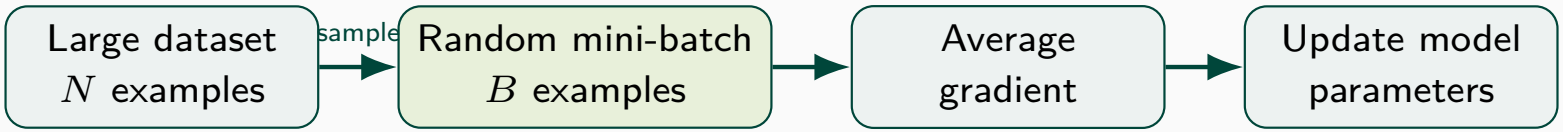
AI Application: Mini-Batch Training

Let the full-data gradient be:

$$\nabla L(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^{N} \nabla \ell_i(\mathbf{w}).$$

A mini-batch estimates it using only B examples:

$$\widehat{\nabla L}(\mathbf{w}) = \frac{1}{B} \sum_{i \in \mathcal{B}} \nabla \ell_i(\mathbf{w}).$$



- Larger batches usually give a lower-variance gradient estimate.
- Smaller batches require less computation per update.
- Batch size therefore creates a variance–computation trade-off.

The Central Limit Theorem

Motivation: What Is the Shape of the Sample Mean?

The LLN tells us:

$$\overline{X}_n \text{ becomes concentrated near } \mu.$$

However, the LLN does not tell us the approximate **shape** of the sampling distribution.

- Is the sampling distribution symmetric?
- How can we calculate
$$P(\overline{X}_n > a)?$$
- How far from μ is a typical sample mean?
- Does the answer depend strongly on the original population shape?

Central Question

Even when the original observations are not Gaussian, can their average have an approximately Gaussian distribution?

The Central Limit Theorem

Central Limit Theorem: Standard Form

Let $X_1, \ldots, X_n$ be i.i.d. random variables with finite mean μ and finite variance $\sigma^2 > 0$. Then:

$$\frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1) \qquad \text{as } n \longrightarrow \infty.$$

Equivalently, for sufficiently large n:

$$\boxed{\overline{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right)}$$

Remarkable Feature

The original population does not need to be Gaussian. Under the stated conditions, the distribution of the **sample mean** becomes approximately Gaussian.

Interpreting the CLT Formula

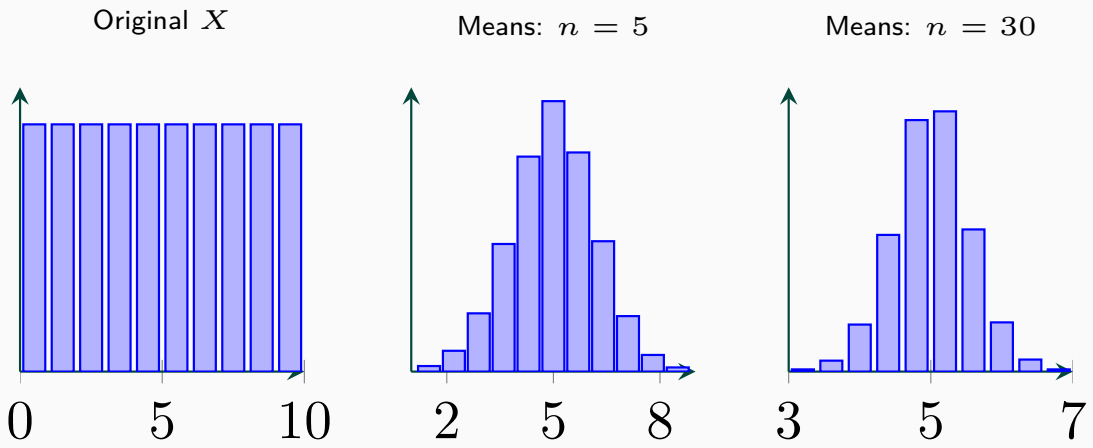
$$\frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} \approx N(0,1)$$

1. Subtract μ:
$$\overline{X}_n - \mu$$
measures the estimation error.
2. Divide by $\sigma/\sqrt{n}$:
$$\frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}}$$
expresses the error in standard-error units.
3. The result is approximately standard normal:
$$Z \approx N(0,1).$$

What Is New?

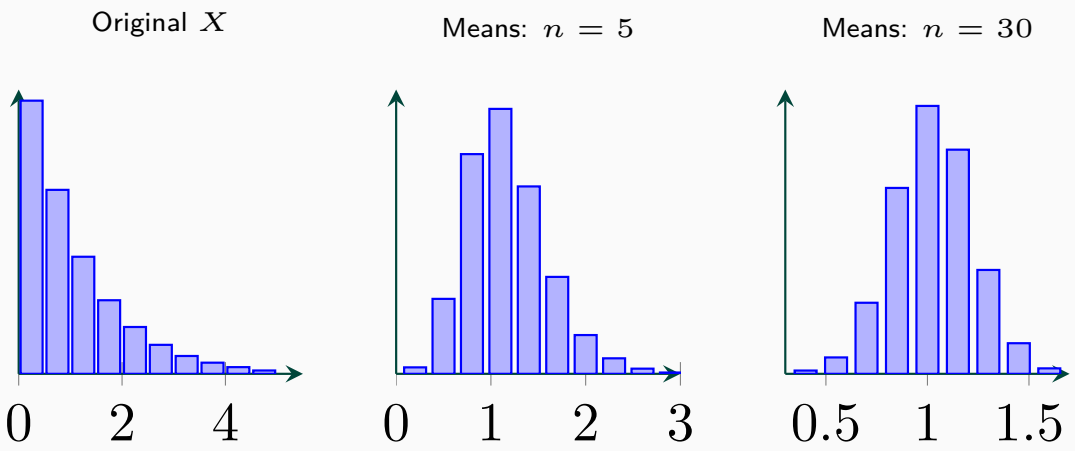
We already knew the centre μ and variance σ^2/n. The CLT additionally tells us that the **shape** is approximately Gaussian.

CLT in Action: Uniform Population



- The original distribution is flat, not Gaussian.
- The distribution of the mean becomes smoother and more bell-shaped.
- Its spread also decreases as n increases.

CLT in Action: Exponential Population



- The original exponential distribution is highly right-skewed.
- Averaging reduces the skewness.
- By $n = 30$, the sampling distribution is much closer to a symmetric bell shape.

Standard Error of the Sample Mean

Since:

$$\mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n},$$

the standard deviation of the sample mean is:

$$\sqrt{\mathrm{Var}(\overline{X}_n)} = \frac{\sigma}{\sqrt{n}}.$$

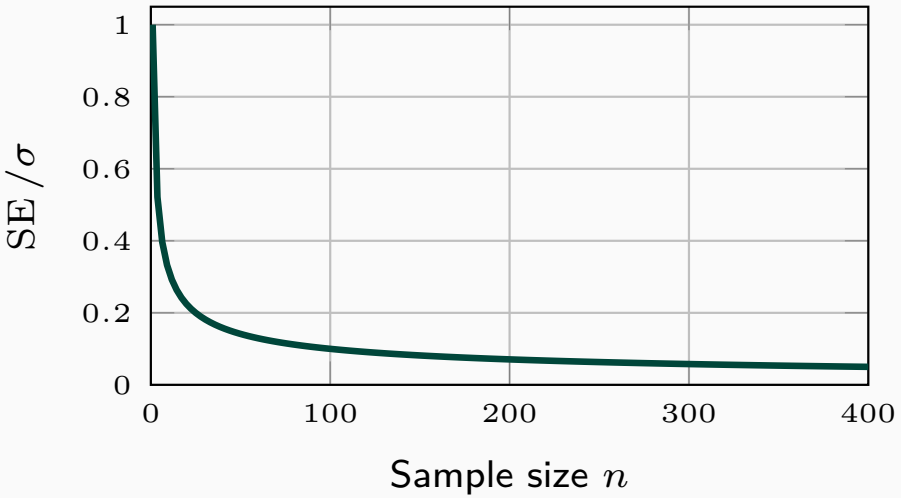
Definition

The **standard error** of the sample mean is:

$$\boxed{\mathrm{SE}(\overline{X}_n) = \frac{\sigma}{\sqrt{n}}}$$

- σ: spread of individual observations.
- $\sigma/\sqrt{n}$: spread of sample means across repeated samples.
- The standard error measures the typical sampling fluctuation of $\overline{X}_n$ around μ.

Sample Size and Standard Error



n	SE
1	σ
4	$\sigma/2$
25	$\sigma/5$
100	$\sigma/10$
400	$\sigma/20$
1000	$\sigma/31.6$

Square-Root Rule

To reduce the standard error by a factor of 2, we need 4 times as much data:

$$\frac{\sigma}{\sqrt{4n}} = \frac{1}{2}\frac{\sigma}{\sqrt{n}}.$$

Worked Example: Applying the CLT

Suppose individual examination scores have:

$$\mu = 70, \qquad \sigma = 12.$$

A random sample of $n = 36$ students is selected. Find:

$$P(\overline{X}_{36} > 72).$$

Worked Example: Applying the CLT

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A random sample of $n = 36$ students is selected. Find:

$$P(\overline{X}_{36} > 72).$$

By the CLT:

$$\overline{X}_{36} \approx N\left(70, \frac{12^2}{36}\right).$$

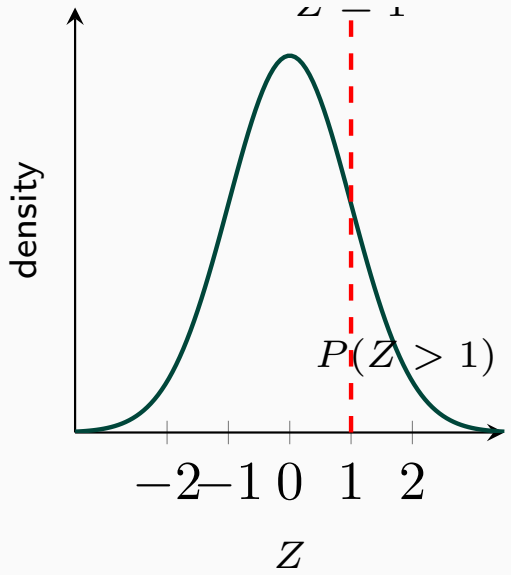
Worked Example: Applying the CLT

Suppose individual examination scores have:

$$\mu = 70, \qquad \sigma = 12.$$

A random sample of $n = 36$ students is selected. Find:

$$P(\overline{X}_{36} > 72).$$



By the CLT:

$$\overline{X}_{36} \approx N\left(70, \frac{12^2}{36}\right).$$

Standardizing:

$$Z = \frac{72 - 70}{2} = 1.$$

Therefore:

$$P(\overline{X}_{36} > 72) \approx P(Z > 1) = 1 - \Phi(1) \approx 0.1587.$$

Common Misconception: What Becomes Gaussian?

Incorrect Statement

"When n is large, the CLT says that each individual observation X_i becomes approximately Gaussian."

This is false.

- Each X_i continues to follow its original distribution.
- Increasing n does not transform an exponential, uniform, or Bernoulli observation into a Gaussian observation.
- The CLT concerns the standardized **sum or sample mean**:

$$\frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}}.$$

Correct Interpretation

The **sampling distribution of the mean** becomes approximately Gaussian as the sample size increases.

LLN and CLT: Putting the Ideas Together

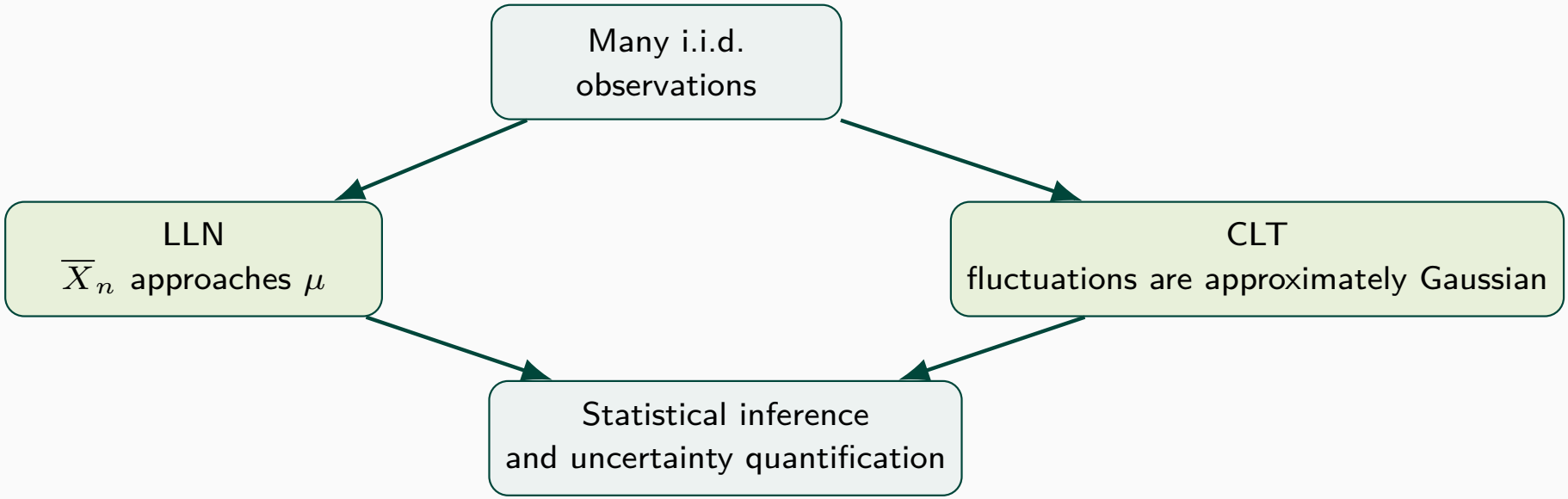
LLN vs. CLT

	Law of Large Numbers	**Central Limit Theorem**
Main question	Does the average approach the population mean?	What is the approximate distribution of the average?
Main result	$\overline{X}_n \to \mu$	$\overline{X}_n \approx N(\mu, \sigma^2/n)$
Main idea	Concentration or consistency	Approximate Gaussian shape
Answers	"Will my estimate become close to the truth?"	"How likely is a particular estimation error?"
Applications	Monte Carlo consistency, stable averages	Probability calculations, confidence intervals, tests

Key Distinction

The LLN concerns where the sample mean goes. The CLT describes the shape and scale of its fluctuations around that destination.

How the Two Theorems Work Together



- The LLN explains why averages can be trusted as estimators.
- The CLT gives a widely used approximation for quantifying their uncertainty.
- Together, they form the foundation of confidence intervals and many hypothesis tests.

Applications in AI and Machine Learning

AI Application: Evaluating Classification Accuracy

Test accuracy is a sample mean. Define:

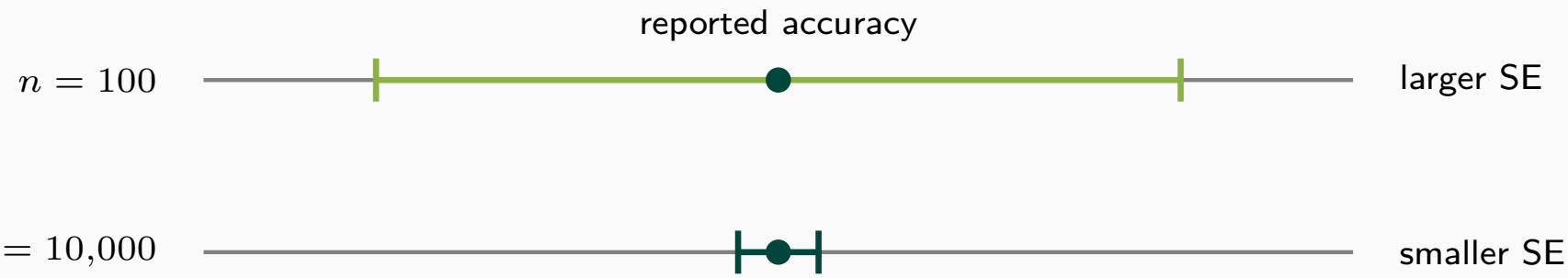
$$X_i = \begin{cases} 1, & \text{if test example } i \text{ is classified correctly,} \\ 0, & \text{otherwise.} \end{cases}$$

Then:

$$\widehat{p} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

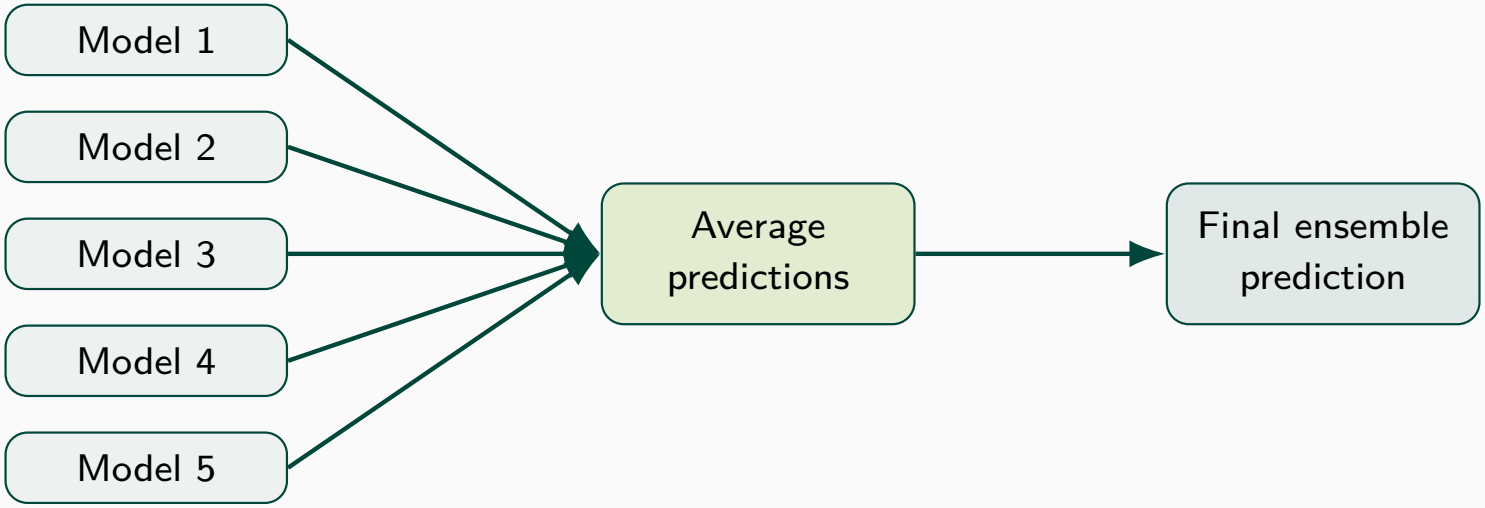
For Bernoulli outcomes:

$$E[\widehat{p}] = p, \qquad \mathrm{SE}(\widehat{p}) = \sqrt{\frac{p(1-p)}{n}}.$$



AI Application: Ensemble Methods and Bagging

Bagging trains several models on random or bootstrapped datasets and averages their predictions.



- Averaging can reduce prediction variance.
- The reduction is strongest when model errors are not perfectly correlated.
- Random Forests use this idea by averaging many randomized decision trees.

Connection

Like a sample mean, an ensemble average tends to be more stable than an individual noisy prediction.

AI Application: Comparing Two Models

Suppose Models A and B are evaluated repeatedly, producing performance measurements:

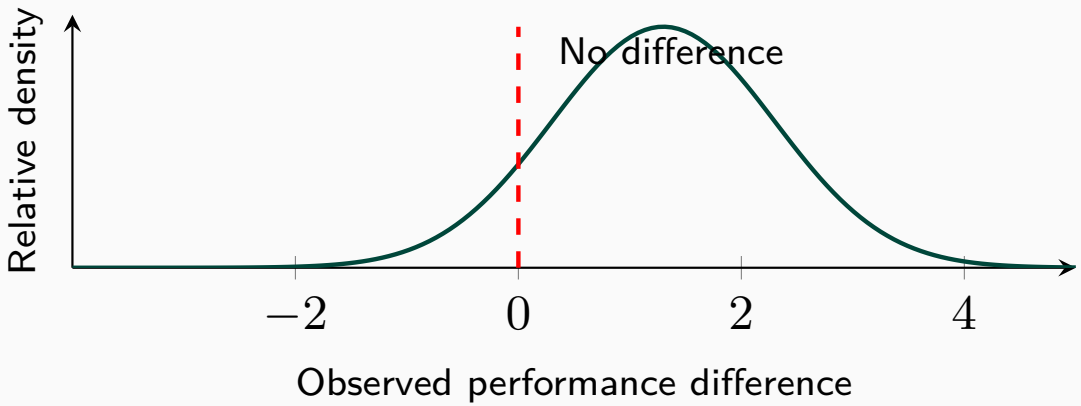
$$A_1, \ldots, A_{n_A}, \qquad B_1, \ldots, B_{n_B}.$$

We are interested in:

$$\overline{A} - \overline{B}.$$

Under suitable independence and sample-size conditions:

$$\overline{A} - \overline{B} \approx N\left(\mu_A - \mu_B,\ \frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B}\right).$$



The CLT helps us judge whether an observed difference is likely to represent a real performance difference or sampling noise.

Summary and Looking Ahead

Summary: Key Formulas

For i.i.d. observations with mean μ and variance σ^2:

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad E[\overline{X}_n] = \mu,$$

$$\mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n}, \qquad \mathrm{SE}(\overline{X}_n) = \frac{\sigma}{\sqrt{n}}.$$

LLN: $\overline{X}_n \longrightarrow \mu$ CLT: $\dfrac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1)$

For large n : $\overline{X}_n \approx N\left(\mu, \dfrac{\sigma^2}{n}\right).$

Common Mistakes (1/2)

- Believing the LLN means short-term outcomes must immediately balance out
- Concluding that the next outcome is affected by previous independent outcomes
- Applying the CLT to individual observations instead of to a sum or sample mean
- Forgetting that the standard error contains $\sqrt{n}$ rather than n:

$$\mathrm{SE} = \frac{\sigma}{\sqrt{n}}$$

Common Mistakes (2/2)

- Confusing the population standard deviation σ with the standard error $\sigma/\sqrt{n}$
- Assuming the original population must already be Gaussian
- Assuming that "large n" has one universal numerical definition
- Ignoring sample size when interpreting a reported mean or model accuracy

Connections Ahead

- Sample mean as an estimator $\longrightarrow$ **Maximum Likelihood Estimation** (Week 8)
- Gaussian sampling distributions $\longrightarrow$ **Confidence Intervals** (Week 11)
- Standardized sample means $\longrightarrow$ **Hypothesis Testing** (Week 12)
- Variance reduction through averaging $\longrightarrow$ **Monte Carlo methods, mini-batches, and ensembles**
- Sampling distributions $\longrightarrow$ **quantifying uncertainty in AI evaluation**

Big Picture

The LLN explains why learning from data is possible; the CLT helps quantify how uncertain our data-based conclusions are.

Class Exercise

Problem

A sensor produces independent readings with population mean $\mu = 50$ and population standard deviation $\sigma = 8$.

A random sample of $n = 64$ readings is collected.

1. Calculate $E[\overline{X}_{64}]$.
2. Calculate $\mathrm{Var}(\overline{X}_{64})$.
3. Calculate the standard error of $\overline{X}_{64}$.
4. Use the CLT to approximate

$$P(\overline{X}_{64} > 52).$$

5. Explain why the LLN alone is not sufficient to answer Part 4. What additional information does the CLT provide?

Work in pairs for 15 minutes, then discuss as a class.

Class Exercise: Solutions

For independent sensor readings with $\mu = 50$, $\sigma = 8$, and $n = 64$:

1. Expected sample mean:
$$E[\overline{X}_{64}] = \mu = \boxed{50}.$$
2. Variance of the sample mean:
$$\mathrm{Var}(\overline{X}_{64}) = \frac{\sigma^2}{n} = \frac{8^2}{64} = \boxed{1}.$$
3. Standard error:
$$\mathrm{SE}(\overline{X}_{64}) = \frac{\sigma}{\sqrt{n}} = \frac{8}{\sqrt{64}} = \boxed{1}.$$
4. By the CLT, $\overline{X}_{64} \approx N(50, 1)$. Hence
$$P(\overline{X}_{64} > 52) \approx P\left(Z > \frac{52 - 50}{1}\right) = 1 - \Phi(2) \approx \boxed{0.0228}.$$
5. The LLN says that $\overline{X}_n$ approaches 50 as n grows, but does not give this probability for $n = 64$. The CLT provides an approximate normal distribution for the sample mean, allowing us to calculate the tail probability.

Preview of Next Week: Maximum Likelihood Estimation

- Today, we assumed that population parameters such as μ and σ^2 were known, and studied how sample averages behave.
- In practice, those parameters are usually unknown.
- Next week, we reverse the question:

  Given observed data, which parameter values best explain the data?
- We will introduce:
  - probability versus likelihood;
  - likelihood and log-likelihood functions;
  - maximum likelihood estimation;
  - Bernoulli parameter estimation;
  - Gaussian mean and variance estimation.

Connection

Maximum likelihood estimation is one of the main mathematical principles behind fitting statistical and machine-learning models.